MIL-S-27178G
17 Dec 82
SUPERSEDING
MIL-S-27178F
20 October 1971

MILITARY SPECIFICATION

SWEEPER, ROTARY, TOWED A/M32U-5A, SNOW, AIR BLAST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercial type of air blast, towed, rotary snow sweeper, designated A/M32U-5A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

VV-F-800	Fuel Oil, Diesel
ZZ-I-550	Inner Tube, Pneumatic Tire

Military

MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-F-3541	Fittings, Lubrication
MIL-H-3992	Hose and Hose Assembly, Rubber, Air and Vacuum Brake, Automatic
MIL-W-5086	Wire, Electric Polyvinyl Chloride-Insulated, Copper or Coppor Alloy Conductor
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-B-11188	Battery, Storage, Lead-Acid
MIL-H-13531	Hose, Rubber and Hose Assembly, Rubber, Hydraulic, Flexible
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMEDO, Robins AFB, GA 31098 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document of by letter.

FSC 3825

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

MIL-F-83002	Fillers, Rotary Sweeper Brush; for Sweeper, Rotary, Towed (A/M32U-5 and A/M32U-5A)

STANDARDS

Federal

H28	Screw Thread Standards for Federal Services
595	Colors

Military

MIL-STD-12	Abbreviations for Use of Drawings, Specifications, Standards, and in Technical Documents
MIL-STD-17	Mechanical Symbols for Aeronautical, Aerospacecraft and Spacecraft Use
MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-209	Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment
MIL-STD-281	Automobile, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS51336	Lunette-Coupler, Drawbar, Ring
MS51317	Light, Warning, Vehicular, Rotating, D.C.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

TIRE AND RIM ASSOCIATION, INCORPORATED (TRA)

Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc, Command Bldg, 34 N Hawkins Ave, Akron, OH 44313.)

DEPARTMENT OF TRANSPORTATION (DOT)

Motor Carrier Safety Regulations

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:03Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The towed sweeper shall be a commercial product of manufacturer's latest design, and shall include the requirements specified herein. It shall be a heavy duty type, with pneumatic tires and a tricycle frame designed to carry a high velocity air blower assembly and an adjustable power driven brush assembly; both assemblies to be mechanically driven by a heavy duty industrial diesel engine. The tricycle frame to be front wheel steer with tow bar and other parts and accessories to make a complete and serviceable unit.

3.2 First article. Unless otherwise specified (see 4.4.1 and 6.2) the contractor shall furnish one or more sweepers for preproduction inspection within the time frame specified (see 6.2) to provide prior to starting production that his production methods and choice of design detail will produce sweepers that comply with the requirements of the specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.2). When specified (see 6.2), the Government will conduct any or all of the first article examinations and tests.

3.3 Components. The sweeper shall consist of the following major components:

Item or System Name	See Requirement
Frame	3.8.1
Power Train	3.8.2
Rotary brush assembly	3.8.3
Blower assembly	3.8.4
Hydraulic system	3.8.5
Electrical system	3.8.6

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.5 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would affect the performance or maintainability of individual components of the overall assembly.

3.5.1 Protective treatment. When materials used in the construction of the sweeper are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

with the performance requirements of this specification. Protective coatings that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall not be used.

3.6 Design and construction. The sweeper shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

3.6.1 Disassembly and assembly. It shall be possible to remove major assemblies with a minimum disconnection of controls, parts, or components. Covers or plates that must be removed for component adjustment or removal shall be equipped with substantial quick-disconnect fastenings.

3.6.2 Reliability. The sweeper shall have a mean-time-between-failures (MTBF) of not less than 217 operating hours demonstrated at 0.9 confidence. For reliability purposes, a failure shall be any failure or malfunction that could prevent satisfactory operation, within specified limits, and safe operation in the primary function of mobile cleaning. Tire failure shall not be considered as reliability failures if the tire loading complies with 3.8.8 through 3.8.8.2.

3.6.3 Maintainability. The sweeper shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.

c. Permit adjustment, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility.

d. Permit maintenance with general purpose tools and equipment normally available commercially. Use of special purpose tools and equipment shall be approved by the procuring activity.

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolt head sizes to the practicable minimum.

f. Foolproofness - Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

Source: https://assist.dla.mil -- Downloaded: 202640-20T11:03Z
Check the source to verify that this is the current version before use.

3.6.3.1 Maintenance downtime. The sweeper shall be designed and constructed to permit maintenance by military service maintenance technicians in not more than the active downtimes specified below. Not more than two technicians shall be required simultaneously for any task.

a. Corrective maintenance downtime - The mean corrective maintenance downtime for all levels of maintenance below depot level maintenance shall be not more than one hour with a related maximum corrective maintenance downtime of not more than three hours.

b. Mean preventive maintenance downtime - Mean preventive maintenance downtime, excluding depot maintenance and major overhaul, shall be not more than one hour.

3.6.4 Flexible hose. Hose for hydraulic lines shall conform to MIL-H-13531, Type II, Grade B. Hose for pneumatic lines shall conform to MIL-H-3992.

3.6.5 Fittings. Disconnect plugs and protected receptacles, multiple line connectors in the electrical system, and readily detachable and attachable type fittings in hydraulic and pneumatic systems, as applicable, shall be utilized wherever practicable to permit rapid component removal and replacement. All disconnect points shall be clearly indicated on the sweeper.

3.6.6 Lifting and tiedown attchments. When specified (see 6.2), lifting and tiedown attachments in accordance with MIL-STD-209, Type III shall be provided to enable the sweeper, less broom, to be lifted in its normal position without damage to the sweeper or its components. The attachments shall be so located that adequate clearance will be provided between lifting slings and all exterior parts of the sweeper when hoisting. Spreader bars will be permitted. Each attachment shall be of sufficient strength to carry the total weight of the sweeper. Markings shall clearly indicate that the attachments are intended for tying down the sweeper.

3.6.7 Safety. All moving parts, or parts subject to high temperature, that are so located as to become a hazard to operating personnel shall be insulated, fully enclosed, or otherwise properly guarded.

3.6.7.1 Work platform. Skidproof work platforms shall be provided in areas where routine work must be performed by personnel standing on the sweeper. The platforms shall be of an open type construction which will permit snow to pass through rather than accumulate.

3.6.7.2 Safety chains. Attached safety chains shall be provided on all removable adjusting and locking pins, caps for quick disconnects, and on all fluid fill caps, except battery. A safety chain and hook arrangement shall be provided to retain the sweeper to the towing vehicle in case of accidental disconnect of towing arm.

3.6.7.3 Sound level. The sound level at the operator's station shall not exceed 85db(A) when measured in accordance with SAEJ919. The exterior sound level shall not exceed 88db(a) when measured in accordance with SAEJ88 (see 4.6.10).

3.6.8 Angle of approach and department. The sweeper, with the brush removed and air nozzle(s) raised, shall have a minimum angle of approach and departure of 20°.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

3.6.9 Lubrication. The sweeper shall be designed and constructed to operate satisfactorily when lubricated with standard lubricants as specified in MIL-M-8090. Lubrication fittings shall conform to MIL-F-3541 and shall be located in acessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication, or which are likely to be overlooked because of inaccessibility. A lubrication chart shall be provided directing attention to all lubricating fittings and shall specify the range and grade of lubricant required for critical temperatures and frequency. The chart shall be permanently attached to the sweeper in an accessible and convenient location.

3.6.9.1 Preservicing. The lubrication and cooling system shall be protected to withstand operation at -40`F. Permanent type antifreeze solution shall be provided. The unit shall be tagged accordingly.

3.7 Performance. The sweeper shall be capable of operating under or withstanding the following conditions without detrimental effect.

3.7.1 Environmental conditions. The sweeper engine shall start within five minutes and all components shall be in full operation within an additional ten minute period under the operating conditions specified below. The sweeper shall satisfactorily withstand the storage conditions specified below without the use of heaters.

a. Operating conditions

(1) Operating temperatures ranging from -25` to +125`F without the use of external heat or installed engine heaters (fluid starting aids may be used for diesel engines.

(2) Operating temperatures ranging from -26` to -40`F with external heat or starting aids.

(3) Altitudes ranging from sea level to 6,000 feet.

(4) All forms of precipitation encountered under field conditions.

b. Storage conditions

(1) Storage temperatures ranging from -40` to +160`F.

(2) All forms of precipitation encountered under field conditions.

3.7.2 Capacity. The sweeper shall efficiently remove two inches of fresh fallen snow (approximately 5 PCF) or reduced depth of proportionately higher density at a rate of not less than 1,584,000 square feet per hour. The sweeper shall be capable of removing 90 percent of loose dry sand spread uniformly at an average density of 0.83 pounds per square foot on a level concrete or equivalent paved surface at least 30 feet long by 12 feet wide at a speed not less than 15 mph.

3.7.3 Mobility. The sweeper shall be capable of being towed in the travel position (broom and air nozzle raised) at not less than 35 mph.

3.7.3.1 Towing force. With the brush in the raised position and the sweeper on dry, level concrete, the force required to pull the sweeper shall be less

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:03Z
Check the source to verify that this is the current version before use.

than 50 pounds per ton of weight.

3.7.4 Endurance. the sweeper shall be capable of performing as described in 3.7.2 for not less than 200 hours without performing any maintenance operations other than lubrication (as published in manuals by the manufacturer) and sweeper broom adjustments. At least 75 percent of the total time shall be for periods of not less than 4 hours continuous operation and at speeds of 25 mph.

3.7.4.1 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or sweeper.

3.8 Details of components.

3.8.1 Frame. The tricycle frame shall be built of wide flanged beams for maximum stiffness and minimum weight and shall be of adequate strength to support the sweeper components when fully equipped for operation under all conditions specified above.

3.8.2 Power train. The power train shall consist of an engine, clutch, drive shafts, brush, and blower drive.

3.8.2.1 Engine. The engine shall be a commercial standard industrial diesel and shall be capable of developing and maintaining sufficient horsepower for efficient operation of all of the sweeper components without exceeding the engine manufacturer's rating for this application. This operation shall be obtained using fuel conforming to VV-F-800. An engine housing with removable side panels shall be provided. The housing and side panels shall be constructed of not less than 16 gauge steel and shall contain an instrument control module, weather proof, with see through covered instrument panel.

3.8.2.1.1 Air cleaner. The engine air inlet shall be provided with an air cleaner of design and capacity as recommended by the engine manufacturer for operating conditions as specified herein. Means shall be provided to prevent snow blowing into and entering the air cleaner intake.

3.8.2.1.2 Fuel system. The fuel system shall be a commercial standard with the fuel tank having sufficient capacity for not less than 8 hours of continuous operation at maximum engine rpm. The tank shall be properly baffled to prevent surging. When terneplate is used, the seams shall be welded so as not to affect the finish inside the tank. The tank shall be so mounted as not to be affected by the engine exhaust heat. The tank shall be equipped with an easily accessible filler neck and drain plug. The filler opening shall permit easy filling from fuel cans or hose. A vented filler cap shall be provided. An accessible fuel shutoff valve shall be provided at the tank outlet if outlet is at bottom of tank.

3.8.2.1.3 Fuel and oil filters. Commercial standard fuel and oil filters shall be provided as recommended by the engine manufacturer for use under conditions as specified herein. The filters shall be readily accessible and shall provide for easy removal and replacement.

3.8.2.2 Clutch. The engine shall be equipped with a clutch of a make and type satisfactory for engagement and disengagement of the rotary brush. The clutch shall be readily accessible for lubricating, adjusting, and replacing of parts. If sealed or lubricated-for-life bearings are employed, lubricants

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

used shall not hinder operation under the temperature conditions specified herein.

3.8.2.3 Drive shafts. The drive shafts shall be commercial standard items suitable for the purpose intended. They shall have suitable guards to prevent them from causing damage to other parts when shear pins break.

3.8.2.4 Brush and blower drive. The brush and blower shall be mechanically driven by the sweeper engine. The type of drive shall be a commercial standard type and adequate for the intended purpose. When the reduction between the engine and the brush or blower is of the chain and sprocket type, it shall be enclosed in an oil-tight case. Adequate means shall be provided for chain take-up to provide for chain and sprocket wear. All chains, sprockets, and gears shall be adequately guarded. The drive engine shall be controllable to provide variable speed of the brush from 450 to 675 RPM. Complete swinging of brush from full 30° right to 30° left or vice versa shall be accomplished in not less than 15 seconds nor more than 30 seconds at engine idle speed.

3.8.3 Rotary brush assembly. The rotary brush assembly shall consist of fillers conforming to MIL-F-83002 and mounting devices mounted on a power driven core assembly. The diameter of the core shall be maintained for the full length around the true center within the specified tolerance of $\pm$ 1/64 inch. The core shall be balanced in accordance with industry standard practice. Provisions shall be made to permit removal of one (or both) broom core end plate(s). Means shall be provided to adjust the broom contact with the paved surface, through not less than 5 inches, to compensate for broom wear and establish broom contact pressures. Raising the broom for transport and lowering the broom for operation shall not change presetting of the brush contact with the paved surface.

3.8.3.1 Brush lock. Provisions shall be made for automatically locking (securing) the brush in the raised position, and the full right, full left, and centered operating positions. Unlocking shall be accomplished by actuating the lowering and turning mechanism controls, respectively.

3.8.3.2 Brush hood. The brush hood shall keep snow and debris picked up by the boom from being thrown back on the sweeper and swept area. It shall be constructed of sheet steel of a thickness suitable for this purpose. The hood shall cover the upper section of the brush and incorporate a front deflector. The deflector shall be hinged to facilitate easy removal of the brush assembly. The hood shall be automatically adjusted to maintain a uniform brush clearance. This adjustment shall be accomplished with the adjustment for ground pressure to compensate for tuft wear (see 3.8.3).

3.8.3.3 Casters. Two casters shall be mounted on the brush frame to support rotary brush assembly and so located that they will always travel in the swept path of the broom. The casters shall be so designed or mounted as to incorporate a closed self compensating hydraulic anti-shimmy system to eliminate caster wheel shimmy and minimize brush bounce during operation. The vertical centerline of the caster kingpins shall remain perpendicular to the ground during brush adjustment (see 3.8.3). The caster wheels shall be free to swivel through horizontal angles of 360° required and encountered in sweeping, turning, and backing operations without interference or tire scuffing. The complete caster assembly shall be easily removable for replacement. The casters shall have pneumatic tires and be suitable for operating under the conditions specified in 3.7.2, 3.7.3 and 3.8.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:03Z
Check the source to verify that this is the current version before use.

3.8.4 Blower assembly. The blower assembly shall be of sufficient capacity to provide an effective discharge of high velocity air, through the nozzle(s) specified below, across the swept path.

3.8.4.1 Air discharge nozzle(s). The air discharge nozzle(s) shall be of a size and so located that the air blast is 90° to the direction of travel and provides a high velocity air blast across the swept path toward the direction of brush discharge. The air blast shall be used to remove loose snow and other loose materials. The capacity and velocity of air discharge shall be sufficient to remove 1/4 inch of water without leaving visible water pockets on a paved surface from the area of starting at the leading edge of the broom path to 8-feet beyond the broom path while traveling at a speed of 25 mph. Provisions shall be made to rotate the air nozzle(s) independently of the brush to permit blowing of snow from runway lights. Features shall be included to assure that full blower discharge, less duct losses, is available at the light during this light cleaning operation.

3.8.5 Hydraulic system. The hydraulic system shall consist of hydraulic cylinders, pump, supply tank, hoses, oil filter, fittings, piping, valves, controls, and all other parts essential to practical and efficient operation of the system. A minimum number of fittings, joints, and connections shall be used to prevent undue backpressure, clogging, and leakage. All hoses passing through metal parts shall be fully insulated to prevent deterimental wear or chafing. The pump shall be driven by the sweeper engine. The pump shall have adequate capacity to provide satisfactory operation of the sweeper. In addition, a manually-operated emergency pump shall be provided and installed. The hydraulic system shall be designed to withstand all loads imposed during operation including loads encountered with heavy accumulation of ice and snow on broom and hood assembly. Under operating conditions as specified herein, the hydraulic system temperature at any point in the system shall not exceed 225°F. All seals, rings, gaskets, et cetera, shall be designed for easy removal and replacement. Armored or otherwise protected rubber or synthetic hose shall be used where flexing is encountered. Provision shall be made for bleeding the system at all necessary points.

3.8.5.1 Hydraulic reservoir. A hydraulic reservoir with a capacity equal to the maximum volume in the hydraulic system plus a minimum of 10 percent additional airspace shall be provided. The reservoir shall have a filler opening of at least 2 inch diameter and a removable strainer in this opening. A shutoff valve and an oil filter that can be easily removed for cleaning/replacement shall be included at the outlet of the reservoir. Provisions shall be made to permit determining the oil level in the reservoir. A placard shall be affixed to the reservoir to show use of, and the system properly filled with, fluid conforming to MIL-H-5606.

3.8.6 Electrical system. The electrical system shall be heavy duty type and adequately weatherproofed. It shall include complete electrical starting, lighting, and an ignition system with an alternator of adequate capacity to provide not less than the sum of the average current of all constant load components plus 35 percent of the average current draw of all intermittent load components. At normal engine idle speed, not less than 7 amp d.c. output shall be provided. The electrical system shall be 24V or 12V potential and and equipped with a transistor type voltage regulator. Two type 6TN batteries per MIL-B-11188 shall be provided. A weatherproof, heavy duty battery box made of steel shall be provided for the batteries. It shall be equipped with an adequately secured, heavy duty lid. Vents shall be provided in the box.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

The box shall be equipped with battery restraints. The box shall be located so that the batteries will be easily accessible for servicing, routine maintenance, removal and replacement. All interior surfaces shall be treated with acid-resistant paint.

3.8.6.1 Lighting system. The sweeper shall be equipped with the following lighting equipment:

a. Taillights, clearance lights, reflectors in accordance with DOT Motor Carrier Safety Regulations 393.14, 393.20 through 393.26(c) and 393.27 through 393.33. The diameter of the reflector and lens shall be not less than 8 inches.

b. At least two floodlights of not less than 50 candlepower each mounted at the front of the sweeper one on each side to light adequately the area being swept and the air discharge nozzle area during runway night cleaning. This lighting shall be effective regardless of sweeping position.

c. Unless otherwise specified (see 6.2) a flasher type warning light, dimensionally conforming to MS 51317, shall be installed on the top of the vehicle. The color shall be aviation yellow as specified in MIL-C-25050. A control switch shall be provided on the instrument panel to control the light. The light assembly shall be mounted for easy removal and installation. The light shall be visible 360° around the sweeper and shall meet the following characteristics:

(1) A flash rate of 60 to 100 flashes per minute.

(2) A minimum light intensity in all vertical planes, expressed in terms of effective candlepower, equal to or exceeding the values as follows:

LIGHT INTENSITIES (With Color Filters)

Angle Above or Below Horizontal in Degrees	Minimum Effective Candle power Light Required
5 above to 5 below	100
5 to 10 above	90
10 to 20 above	70
20 to 30 above	50

3.8.6.2 Wiring.

3.8.6.2.1 Wire. The wire used in the electrical system shall conform to MIL-W-5086 and shall be encased, when applicable, in a flexible, nonmetallic tubing conforming to MIL-I-7444. The wire shall be so installed as to receive the most protection from other components of the sweeper.

3.8.6.2.2 Circuit marking. The circuits on the sweeper shall be color coded or numbered, and so indicated on the wiring diagram.

3.8.6.2.3 Grommets and clamps. Grommets or clamps shall prevent chafing where the wire passes through structural members.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:03Z
Check the source to verify that this is the current version before use.

3.8.6.2.4 Protective safety devices. Protective devices that will automatically shut off the engine shall be installed in the systems to protect the engine and the hydraulic system in event of any of the following malfunctions:

a. Hydraulic oil temperature return to reservoir exceeds 225`.

b. Vacuum in hydraulic suction line exceeds 4 inches of HG.

c. Hydraulic reservoir oil level falls below recommended level.

d. Engine coolant temperature exceeds 205`F.

e. Engine lubricating oil pressure falls below 4 PSI.

In the event any of the above malfunctions occur, manually reset relays with indicator lights will automatically shut down the engine and indicate which system is malfunctioning. The relays for this system shall be installed in the electrical instrument control module located in the engine housing.

Self acting check valves shall be installed in the hydraulic oil return lines before the reservoir to prevent oil spillage on runway in case of line breakage.

3.8.6.3 Electrical instrument and control module. The electrical control module shall be weatherproof and readily removable by utilizing multiple contact quick disconnect plugs. The module shall contain the following engine instruments and controls, ammeter, engine oil pressure gauge, coolant temperature gauge, fuel gauge, hourmeter, tachometer, ignition switch, and starter button. In addition to the engine instruments and controls, the module shall contain all electrical components such as relays, vacuum switches, and safety switches as related to the hydraulic system. The module shall be installed in such a manner that if an electrical problem does occur, it can be removed and a new module installed in less than 15 minutes.

3.8.7 Towbar. The towbar shall be designed for towing by vehicles having a pintle hook height between 30 and 40 inches above ground level. The towbar shall be equipped with a lunette eye conforming to MS 51336. Safety chains shall be provided.

3.8.8 Pneumatic tires. Unless otherwise specified (see 6.2), tires shall be either tube or tubeless type with manufacturer's standard commercial tread. Tires shall not be loaded in excess of the specified load limits in accordance with TRA Yearbook (see 3.6.2). Tires shall not be less than 100 level quality. Manufacturer's current size of tires conforming to specified requirements may be furnished.

3.8.8.1 Wheels and rims. The sweeper shall be equipped with at least three basic wheels (exclusive of broom casters), one front and two rear wheels, of the tricycle type. The front wheel assembly shall be attached to the towbar and swivel mounted in a manner that will enable the wheel to be changed without exposing the wheel bearings. Wheel bearings shall be sealed to prevent loss of lubricant and entrance of moisture and abrasives. The front wheel shall be so positioned that debris from sweeping is not thrown onto the wheel during sweeping operations to either side. Wheels shall be the manufacturer's current commercial standard type. Rims shall conform to the TRA recommendations for the type and size of tire furnished.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the currentversion before use.

3.8.8.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy duty type in accordance with ZZ-I-550 and shall be of the proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with TRA recommendations.

3.8.9 Remote controls. The operation of the sweepers, except for engine starting, shall be remotely controlled from the cab of the towing vehicle. The controls shall be remotely actuated by controls mounted on a control panel or box. The panel or box shall have a handle and shall be so attached to a flexible electric cable that it can be easily placed or removed from the cab. The cables shall be enclosed in a flexible, weatherproofed, rubber or neoprene conduit and extend at least 25 feet ahead of the lunette eye of the sweeper. Suitable hooks shall be provided for storing the cable assembly. The size of the control panel or box shall be kept to a minimum for easy handling. The panel or box shall be fully waterproofed and shall incorporate an emergency engine cutoff pushbutton, a warning buzzer to indicate termination of hydraulic operation of brush and air nozzles positioning. A tachometer shall be provided to indicate the brush RPM. A fourway toggle switch for controlling the valves shall be of the heavy-duty type and suitably marked for each of the following operations:

a. Raising and lowering brush and air nozzle(s).

b. Revolving brush and air nozzle(s), left and right.

c. Revolving air nozzle(s), only for cleaning lights left and right.

d. A two-way toggle switch to govern engine at idle and full speed.

3.8.9.1 Hydraulic valve compartment. The hydraulic valve compartment shall be completely enclosed with easy access doors. One copy each of the wiring and hydraulic diagram shall be securely fastened to the inside of the compartment cover. Drain holes shall be provided in the lowest point of the compartment. The valve controls shall be clearly marked. Control circuits shall extend from the valve compartment to the forward end of the sweeper and then to the remote control box specified in 3.8.9.

3.8.9.2 Safety device. A safety device shall be utilized that will require the engine to be at idle speed when the operator engages the clutch.

3.9 Part numbering of interchangeable parts. All parts having the same manufactorer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.10 Electromagnetic interference. The sweeper shall be so constructed and shielded that electromagnetic interference generated during operation under any condition specified herein will not exceed the limits established in MIL-STD-461, Class D3.

3.11 Dimensions. The overall road dimensions of the sweeper, exluding the rotary brush, shall not exceed 130 inches in width and 100 inches in height. When the broom and nozzle assemblies are in the raised position, the ground clearance shall be not less than 8 inches for the nozzle and 4 inches for the broom.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

3.12 Weight. The sweeper shall be of the minimum weight consistent with the requirements specified herein.

3.13 Threads. All screw and pipe threads shall conform to FED-STD-H28.

3.14 Treatment and painting. The sweeper shall be treated and painted in accordance with the manufacturer's standard commercial practice. The color, except the engine, shall be yellow, color 13538, conforming to Standard 595.

3.15 Markings.

3.15.1 Transportation data plates. Transportation data plates marked and sized in accordance with MIL-P-514 shall be provided. All marking plates, placards, charts, instruction plates, et cetera, shall conform to MIL-P-514. Unless otherwise specified, the size, shape, and location shall be consistent with information required thereon. Abbreviations shall be in accordance with MIL-STD-12. Symbols shall be in accordance with MIL-STD-17.

3.15.2 Electromagnetic interference instruction plate. The following information shall be provided on a plate installed in a conspicuous location on the engine instrument panel:

CAUTION

This vehicle is equipped with radio interference suppression devices. Care should be exercised during vehicle maintenance to retain and reinstall these devices prior to reuse of the vehicle.

3.15.3 Tire pressure. The tire inflation pressure shall be stenciled in black as near each wheel as practicable.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship. The sweeper, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, weld spatter, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring, riveting, alignment of parts, tightness of assembly screws and bolts, and other requirements as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the sweeper

Source: https://assist.dla.mil -- Downloaded: 20260-20T11:03Z
Check the source to verify that this is the current version before use.

shall be classified as follows:

a. First Article testing (see 4.4).

b. Acceptance tests (see 4.5).

4.3 Test condition. All sweeping tests shall be performed within an ambient temperature range of -25° to +125°F. The requirements of each individual sweeping test specified herein shall be met by one sweeper.

4.4 First Article testing (see 6.2).

4.4.1 Test sample. Unless otherwise specified (see 6.2), the preproduction test sample shall consist of one sweeper representative of the production equipment. It shall be subjected to the tests specified in 4.6. Presence of one or more defects shall be cause for rejection. Failure of any test shall be cause for rejection.

4.5 Acceptance test. The acceptance tests shall consist of the folowing:

a. Individual test.

b. Sampling plan and test.

4.5.1 Individual test. Each sweeper shall be subjected to an examination of product test as described in 4.6.1.

4.5.2 Sampling plan and test.

4.5.2.1 Periodic sampling. One sweeper shall be selected at random from every 50 or fraction thereof produced and subjected to the sweeping performance test as described under 4.6.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or produced later shall be accepted until the extent and cause of failure have been determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The sweeper shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking. Any deviation from these requirements shall be cause for rejection. For the preproduction item, this examination shall be accomplished by using a checklist which lists each requirement in Section 3 that is not validated by tests in Section 4. The results of this examination shall be noted beside each requirement and enclosed as part of the test report.

4.6.2 Environmental tests. The sweeper shall be subjected to the following environmental tests in accordance with MIL-STD-810.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:03Z
Check the source to verify that this is the current version before use.

4.6.2.1 High temperature. The sweeper shall be subjected to high temperature in accordance with method 501, procedure 1. The operating temperature after the exposure period shall be 125°F. The sweeper shall be operated for one hour while maintaining the broom rpm and contact area determined necessary to meet the sweeper performance requirements of 4.6.3. Bristle breakage, overheating of the engine or hydraulic system, or deterioration of component parts shall not be evident during this test.

4.6.2.2 Low temperature. The sweeper shall be subjected to low temperature in accordance with method 502, procedure 1. The operating temperature after the exposure period shall be -25°F. The sweeper shall be started and operated for one hour while maintaining the broom rpm and contact area and blower output determined necessary to meet the sweeper performance requirements of 4.6.3. The engine and operating components shall be started and engaged respectively without the use of external heat sources. Bristle breakage shall not be evident during this test. The operating temperature shall be changed and stabilized at -40°F. The sweeper shall be started and operated as described above. Starting aids and external heat may be used.

4.6.3 Sweeping performance. Sweeping performance test as described in MIL-F-83002 shall be conducted. Sand removal, as described in the performance test of that specification shall be not less than 90 percent with the blower assistance.

4.6.4 Endurance. The sweeper shall be operated in a sweeping position as described in the filler life test of MIL-F-83002. During this test, there shall be no evidence of excessive brush bounce or caster shimmy. (Excessive bounce shall be evidenced by failure to maintain clean pavement over the entire swept area.)

4.6.4.1 Adjustments. During the endurance test, only lubrication and adjustments of the brush to compensate for wear shall be permitted.

4.6.5 Mobility. The sweeper with broom and nozzle in the raised position shall be towed at speeds ranging from 25 to 35 mph over improved roads or runways for not less than 500 miles without damage.

4.6.5.1 Towing. The force required to tow the sweeper on dry, level concrete, with the brush in the raised position, shall be measured and shall not exceed 50 pounds per ton of weight.

4.6.5.2 Angle of approach and departure. The sweeper, with the brush removed and air nozzle(s) raised, shall enter, negotiate, and leave a ramp which has a slope of 20° without scraping. The length of the ramp shall be not less than the overall length of the sweeper.

4.6.6 Electromagnetic interference. The sweeper shall be tested as a complete unit for electromagnetic interference in accordance with MIL-STD-461, Class D3, and MIL-STD-462.

4.6.7 Reliability demonstration. Compliance with the quantitative reliability requirements shall be considered demonstrated when all tests specified herein have been satisfactorily completed with no failure preventing satisfactory or safe operation in the primary function of mobile cleaning. Normal adjustments and servicing as specified in 4.6.4.1 and elsewhere herein need not be counted as reliability failures. Apparent failures that are caused

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

ment failure) need not be counted as reliability failures if the cause can be substantiated to the satisfaction of the procuring activity. Any test resulting in a failure shall be documented. Acceptance for reliability purposes shall not be made until, to the satisfaction of the procuring activity, each failure has been analyzed, the cause determined, design or other modifications made to preclude repetition of the failure or related failures, successful retest made by the contractor of the failed equipment with the modifications included, and contractor action taken to reflect the modifications in equipments to be delivered. Any retest shall include all tests, in addition to a particular failed test that could be influenced by the modification.

4.6.8 Maintainability demonstration. Compliance with qualitative maintainability requirements shall be demonstrated by inspection and examination. Compliance with the quantitative maintenance downtime requirements shall be demonstrated task performance as specified below. The active maintenance downtime, the tools and equipment required, the skills required, and any obstructions or conditions hindering the task or presenting potential hazards to personnel or to parts shall be observed and recorded. Signs of excessive wear or other indication of potential failure shall be recorded where observed and the information supplied to the engineering activity for study and for recommendation of any appropriate modifications to be reflected in future items delivered. Such recommendations shall be approved by the procuring activity before implemented or successful retest or other tests specified herein may be required where the test result could be influenced by the modification. It shall be demonstrated that the sweeper is in satisfactory operating condition after the maintenance tasks have been done. The frequency of servicing required for the various components shall be determined by study of existing suppliers' recommendations and by servicing required during the preproduction tests. Any tendency of the sweeper to pick up mud, snow, or ice which could interfere with servicing, operation or maintenance shall be noted.

a. Corrective maintenance. Compliance with the quantitative corrective maintenance downtime requirements shall be demonstrated by a maintenance demonstration procedure, maintenance task selection, and maintenance task performances in accordance with MIL-STD-471 including appendixes A and B, method 1, test plan pair (A1 plus B1).

b. Preventive maintance. Compliance with the quantitative preventive maintenance downtime requirement shall be demonstrated in accordance with MIL-STD-471, appendix B, method 6. Each task need not be done more than once, but the resulting times shall be weighed in mean-time calculations to account for the expected frequency of each task. Tasks termed servicing tasks shall be included in this demonstration. Repetitive subtasks, such as removing panels, common to several tasks need not be repeated for each related task if the time established for the subtask is included in the recorded total time for each related task. The task of replacing the tufts in the brush assembly need not be included as part of the preventive maintenance demonstration.

4.6.9 Sound Level Test. The sound level at the operator's station and at the vehicle exterior shall be measured in accordance with SAEJ919 and SAEJ88 respectively. Sound levels shall not exceed requirements in 3.6.7.3 Sound levels.

4.6.10 Inspection after test. After the completion of testing, a thorough inspection of the sweeper shall be accomplished as follows. Any fault(s)

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

shall constitute failure of the test.

a. Major assemblies, such as engine, clutch(s), axles, and drive trains, shall be disassembled as necessary to determine the extent of wear and damage. Failures, erratic operation, excessive noise, or vibration suggesting potential failures shall be indicative that disassembly is necessary.

b. All gears, sprockets, shafts, bearings, parts, and components of assemblies which are disassembled shall be subjected to magnetic particle inspection to determine if cracks have developed.

c. Castings used to house major assemblies or used as part of major assemblies which are disassembles shall be examined under black light to determine if damage was incurred during testing.

4.7 Lifting and tiedown attachment tests. When required, the lifting and tiedown attachments shall be tested to conform to 3.6.6.

4.8 Servicing and adjusting. Prior to acceptance of the vehicle(s) by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: Focusing of lights; adjustment of engine(s); adjustment of electrical system; filling and charging of batteries (except where dry charged batteries are specified); front end alignment (steering); inflation of all tires; complete lubrication of running gear components and chassis (frame) including engine(s) and mounted equipment with the grades of lubricants recommended for the ambient temperature at the destination point; servicing of cooling system(s) with a solution of ethylene glycol and water in equal parts by volume (see 3.6.9.1).

4.9 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.2).

5.1.1 Level A. The sweeper shall be preserved and packaged in accordance with MIL-STD-281.

5.1.1.2 Tools. When tools are furnished, they shall be securely wrapped and packed within the storage compartment with packaging and cushioning materials conforming to MIL-STD-1186.

5.1.2 Level C. The sweeper shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing (see 6.2).

5.2.1 Levels A and B. Unless otherwise specified (see 6.2), sweepers preserved and packaged as specified in 5.1.1 shall be shipped mobile in accordance with MIL-STD-281.

5.2.2 Level C. Packages which require overpacking shall be packed to insure carrier acceptance and safe delivery to destination in compliance with Official Classification Committee, Uniform Freight Classification Rules, American Truck-

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

in Association, National Freight Classification Rules or with others applicable to the mode of transportation.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

SWEEPER, ROTARY, TOWED, A/M32U-5A, SNOW, AIR BLAST

6.1 Intended use. The A/M32U-5A sweeper is intended for use in the removal of new fallen snow, ice or new snow residue, and all season sweeping of runways, taxiways, and ramp areas, and other airfield paved areas.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When lifting attachments are required (see 3.6.6).

c. When type warning lights required are other than specified (see 3.8.6.1c).

d. Registration number required.

e. When type tires required are other than specified (see 3.8.8).

f. Location and condition for preproduction testing (see 4.4).

g. When more than one preproduction sample is required (see 4.4.1).

h. Applicable level of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Definition.

6.3.1 MTBF. For reliability purposes, MTBF shall be the average (arithmetic mean) operating time between failures that prevent satisfactory or safe performance of primary functions. Note that MTBF demonstrated at a stated confidence level will be less than a point estimate derived by deviding the accumulated operating time by the number of failures.

Custodian:
Air Force - 99
Army - ME
Navy - YD

User:
Army - CE
Air Force - 11

Preparing Activity:
Air Force - 84

Project Number:
3825-0147

Source: https://assist.dla.mil -- Downloaded: 20260-20T11:03Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

HQ, AFLC CASO/LODSAC
Federal Center
Battle Creek, MI 49016

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-27178G

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   [ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:03Z
Check the source to verify that this is the current version before use.